Gyo-Jin Kang

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Technical Skills

Languages: Python, SQL, R, Java, JavaScript, HTML/CSS, Bash
Frameworks/Libraries: PyTorch, TensorFlow, Scikit-Learn, LangChain, FastAPI, dbt, Pandas, NumPy, GCP, React
Tools/Platforms: Snowflake, Docker, Kubernetes, AWS (EKS/ECR), GitHub Actions, Git/GitHub, ArgoCD

Experience

Data Scientist — Jan 2026 – Aug 2026
Workstream — *Burnaby, BC*

- Built a multi-agent legal-compliance platform for QSR franchises, using incremental **dbt** models and automated CI checks to flag potential statutory violations and cut manual compliance review time by **73%**. Developed an intuitive frontend design using **Typescript** and **React** to enhance user engagement.
- Developed a **FastAPI** + **LangChain** Text-to-SQL agent backed by custom **MCP (Model Context Protocol)** servers and embedding-based retrieval over **Snowflake**, reducing ad-hoc analytics requests by **30%**.
- Built a self-evolving **LLM-as-a-Judge** evaluation loop that scores agent responses against a rubric and auto-refines its own prompts/references, catching regressions pre-release and reducing manual QA.
- Containerized and deployed the agent suite with **Docker** and **Kubernetes** on **AWS EKS**, using **GitHub Actions** CI/CD to automate build, test, and rollout.
- Built and maintained **ELT** pipelines landing data into **Snowflake** — managing **Hevo** ingestion across 5+ SaaS and database sources, and hand-building a custom ELT pipeline for **Datadog** to capture mobile app performance.

Data Analyst — Jan 2025 – Aug 2025
British Columbia Lottery Corporation — *Vancouver, BC*

- Designed and ran **A/B tests** with power-analysis-driven sample sizing to optimize slot return-to-player (RTP) configurations; the winning variant lifted digital GGR by **8%**.
- Developed an **Isolation Forest** anomaly-detection service to streamlining automated fraud review.
- Built player segmentation (**Gaussian Mixture Models**) and uplift models (**X-Learner**, **CATE**) to target high-responsiveness players, informing campaigns credited with **$1M+** incremental Q4 revenue.

Machine Learning Engineer (Part-Time) — Sep 2024 – Jul 2025
UBC MINT — *Vancouver, BC*

- Built a **CNN-LSTM** hybrid model for EEG motor-imagery decoding, reaching **82%** classification accuracy through optimized spatio-temporal feature extraction.
- Engineered a robust feature extraction pipeline using **Riemannian geometry** (signal covariance + tangent-space mapping, per *Barachant et al.*) to keep EEG decoding accurate despite noise and session-to-session variability.
- Prevented data leakage by enforcing strict causal temporal splitting between train and test windows, enabling realistic offline evaluation on noisy EEG signals.

Projects

Breast Cancer Segmentation | *TensorFlow* — August 2024

- Built a **U-Net** model to segment malignant tumor regions in breast ultrasound images.
- Implemented an encoder–decoder architecture with transposed-convolution up-sampling, skip connections, batch normalization, and dropout; evaluated with the Dice coefficient.

Education & Certification

University of British Columbia — Vancouver, BC
Bachelor of Science in Statistics; Economics Concentration — *Sep. 2022 – May 2027 (Expected)*

Deep Learning Specialization — Coursera
CNNs, RNNs, and Transformers with modern optimization and regularization techniques — *Mar. 2025*